MICHAEL S YAO michaelsyao.com michael.yao@pennmedicine.upenn.edu Google Scholar

OVERVIEW

I am a physician-scientist in-training at the University of Pennsylvania. My research focuses on how we can build trustworthy, evidence-based ML systems for healthcare. I leverage prior knowledge and statistical methods to adapt algorithms to different patient populations, with applications in rare disease diagnostics, clinical decision making, and precision medicine.

EDUCATION

MD-PhD Bioengineering, University of Pennsylvania 2021 - Present
NIH F30 NRSA Fellow | HHMI-NIBIB T32 Fellow | Medical Education Concentration
Advised by Osbert Bastani and James Gee | PhD Thesis Link | Research Award | Step 2 CK 260

MS Computer Science, University of Pennsylvania 2023 - 2025
Penn Engineering Teaching Award

BS Applied Physics, California Institute of Technology 2017 - 2021
Salutatorian | Full-Ride Merit Scholarship Award | Advised by Mikhail Shapiro

EXPERIENCE

ML Scientist, **Genentech** San Francisco, CA | 2025
Designed algorithms for optimizing personalized treatment strategies and efficient patient-clinical trial matching using large language models (LLMs).

AI Research Consultant, Scale AI Remote | 2025
Engineered medical reasoning benchmarks for evaluating the clinical capabilities of frontier LLMs. Critiqued and red-teamed medical usage of proprietary LLMs prior to public use.

AI Clinical Fellow, **Glass Health** Remote | 2023-2024
Authored and assessed clinical guideline articles as knowledge sources for LLM-based clinical decision support tools. Investigated applications of LLMs for medical education.

PhD Research Intern, **Microsoft Research** Redmond, WA | 2022
Developed ML methods for accelerated MRI imaging and improving the generalizability of MRI image reconstruction models to different anatomies and patient populations.

Software Engineer, **Hyperfine** Guilford, CT | 2021
Implemented and validated algorithms for more robust MRI signal acquisition and image post-processing in MR software across 25+ hospital sites.

MCAT Content Author, Blueprint Test Prep Remote | 2020-2022
Authored practice questions and educational material for MCAT test preparation offerings.

SELECTED WORKS

[1] **Yao MS**, Bastani O, Andersson A, Biancalani T, Bentaieb A, Iriondo C. Knowledgeable language models as black-box optimizers for personalized medicine. **ICLR**. (2025). Link

[2] **Yao MS**, Chae A, Saraiya P, Kahn CE, Witschey WR, Gee JC, Sagreiya H, Bastani O. Evaluating acute image ordering for real-world patient cases via language alignment with radiological guidelines. **Nat Commun Med**. (2025). Link | Penn News | Research Award

[3] **Yao MS**, Huang L, Leventhal E, Sun C, Stephen SJ, Liou L. Leveraging datathons to teach AI in undergraduate medical education. **JMIR Medical Education**. (2025). Link
[4] **Yao MS**, Gee JC, Bastani O. Diversity by design: Leveraging distribution matching for offline model-based optimization. **ICML**. (2025). Link
[5] **Yao MS**, Zeng Y, Bastani H, Gardner J, Gee JC, Bastani O. Generative adversarial model-based optimization via source critic regularization. **NeurIPS**. (2024). Link
[6] Wu Y, Liu Y, Yang Y, **Yao MS**, Yang W, Shi X, Yang L, Li D, Liu Y, Gee JC, Yang X, Wei W, Gu S. A concept-based interpretable model for the diagnosis of choroid neoplasias using multimodal data. **Nature Communications**. (2025). Link
[7] Yang Y, Gandhi M, Wang Y, Wu Y, **Yao MS**, Callison-Burch C, Gee JC, Yatskar M. A textbook remedy for domain shifts: Knowledge priors for medical image analysis. **NeurIPS (Spotlight)**. (2024). Link
[8] **Yao MS**†, Chae A†, MacLean MT, Verma A, Duda J, Gee JC, Torigian DA, Rader D, Kahn CE, Witschey WR, Sagreiya H. SynthA1c: Towards clinically interpretable patient representations for diabetes risk stratification. **Prime MICCAI**. (2023). Link

TEACHING

Instructor and Curriculum Lead, **Ethical Algorithms for the Modern Clinician** | Link
Head TA, **Pre-Clinical Medicine** (Clerkship Prep, Penn) | 2026, 2023
Head TA, **Healthcare and Technology** (CIS 7000, Penn) | 2026, 2024, 2023
TA, **Distributed Computer Systems** (CIS 5050, Penn) | 2026, 2025
TA, **Principles of Deep Learning** (ESE 5460, Penn) | 2024
TA, **Imaging Informatics** (EAS 5850, Penn) | 2024
TA, **Diagnostic Ultrasound for Medical Students** (Penn) | 2024, 2023

Head TA, **Applied Mathematics** (ACM 95ab, Caltech) | 2021, 2020
TA, **Graduate Classical Physics** (Ph 106a, Caltech) | 2020
TA, **Quantum Physics** (Ph 12ab, Caltech) | 2020, 2019
TA, **Operating Systems** (CS 24, Caltech) | 2019
TA, **Special Relativity and Electromagnetism** (Ph 1bc, Caltech) | 2019, 2019

SERVICE

Organizations
Co-President, Penn HealthX Student Group
Co-President, Medical Education Club, University of Pennsylvania
Technology Committee Vice-Chair, American Physician Scientists Association
Board Member, Medical Education Club, University of Pennsylvania
AI Curriculum Task Force Member, University of Pennsylvania
Admissions Committee, University of Pennsylvania
Director of Data Science & AI, MDplus

Editor-in-Chief, Caltech Undergraduate Research Journal
Volunteer Tutor, Caltech RISE Tutoring Program
Peer Tutor, Caltech Deans' Office
Student Body Representative, Caltech Academics and Research Committee

Referee
AMIA Annual Symposium
npj Digital Medicine
Neural Information Processing Systems (NeurIPS)
International Conference on Learning Representations (ICLR)
International Conference on Machine Learning (ICML)
AAAI Conference on Artificial Intelligence
AHLI Conference on Health, Inference, and Learning (CHIL)

Invited Talks
ML for Clinical Medicine and Education, Tufts University School of Medicine | 2026
Fueling Reimagination to Advance Medical Education, University of Pennsylvania | 2026

INVITED COMMENTARY

[9] **Yao MS**, Chae A. Refining the second reader: Artificial intelligence as a triage partner in population-based breast cancer screening. **Radiology: AI** (Editorial). (2026). Link

[10] Gee JC, **Yao MS**. Effective structured information extraction from chest radiography reports using open-weights large language models. **Radiology** (Editorial). (2025). Link

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ADDITIONAL WORKS

[11] **Yao MS**. Large language model-based evaluation of the impact of gender in medical research. Preprint. (2026). Link

[12] Barinaga Z, Chae A, Hashimoto DA, Jiang J, Leventhal E, Meng S, Purcell M, **Yao MS**. Ethical algorithms for the modern clinician. **NeurIPS Education Program**. (2025). Link

[13] Meng S, Purcell M, **Yao MS**, Hashimoto D. Implementing a novel artificial intelligence in medicine elective curriculum. **AAMC Annual Meeting** (Poster). (2025).

[14] Chae A†, **Yao MS**†, Sagreiya H, Goldberg AD, Chatterjee N, MacLean MT, Duda J, Elahi A, Borthakur A, Ritchie MD, Rader D, Kahn CE, Witschey WR, Gee JC. Strategies for implementing machine learning algorithms in the clinical practice of radiology. **Radiology**. (2024). Link

[15] **Yao MS**, Hansen MS. A path towards clinical adaptation of accelerated MRI. **Machine Learning for Health**. (2022). Link

[16] Abedi MH†, **Yao MS**†, Mittelstein DR, Bar-Zion A, Swift MB, Lee-Gosselin A, Barturen-Larrea P, Buss MT, Shapiro MG. Ultrasound-controllable engineered bacteria for cancer immunotherapy. **Nature Communications**. (2022). Link | Patent

[17] **Yao MS**, Uhr L, Daghlian G, Amrute JM, Deshpande R, Mathews B, Patel SA, Henri R, Liu G, Reierson K, Johnson G. Demonstration of a longitudinal medical education model to teach point-of-care ultrasound in resource-limited settings. **POCUS J**. (2020). Link

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PERSONAL PROJECTS

Medical Education: Ethical Algorithms for the Modern Clinician | Clerkship Connections
Pimping Rounds | Substack
Computer Programming: Advent of Code 2025 **|** Project Euler Puzzles | NRMP Match